

Unsupervised Learning Finding Hidden Groups

What is Unsupervised Learning?

In supervised learning you have a label (the answer) for every row of training data. In unsupervised learning there are no labels — the algorithm must discover structure, patterns, and groups entirely on its own.

Supervised Learning vs Unsupervised Learning

Feature	Supervised Learning	
Labels	Requires labeled data (target column available)	No labels required
Goal	Learn from known answers and make predictions	Discover hidden patterns
Input Data	Features + Target (X and y)	Features only
Output	Predicted class or numerical value	Clusters, centroids
Common Algorithms	Logistic Regression, Linear Regression, Decision Tree, Random Forest	K-Means, PCA
Evaluation Metrics	Accuracy, Precision, Recall, F1-Score, ROC-AUC, RMSE	Silhouette Score
Teacher Present?	Yes (correct answers available)	No (model learns from data)
Typical Questions	"Will this customer buy?" "Is this email spam?"	"Can customers be grouped?"
Real-World Examples	Spam Detection, Disease Prediction, House Price Prediction, Fraud Detection	Customer Segmentation

Few examples where we can use non-supervised learning.

Imagine you are given a large dataset, but there are no answers, labels, or target column provided.

For example:

A shopping website has data about 1 lakh customers.

A news website has 10,000 articles.

A bank has millions of transactions.

Nobody tells us:

Which customer belongs to which group.

Which articles belong to which topic.

Which transactions are fraudulent.

The goal is to let the machine discover hidden patterns automatically.

This is called Unsupervised Learning.

Supervised vs Unsupervised Side by Side

See the difference in how you set up each type of problem:

```
# ■ Supervised vs Unsupervised Comparison ■
import numpy as np, pandas as pd, matplotlib.pyplot as plt
from sklearn.datasets import make_blobs

# Generate toy data: 300 points, 3 natural clusters
X, y_true = make_blobs(
    n_samples=300,
    centers=3,
    cluster_std=1.2,
    random_state=42)

# X → feature matrix (coordinates)
# y_true → the TRUE labels (we'd have these in supervised learning)
# ■ Supervised approach: pretend we have labels ■
from sklearn.linear_model import LogisticRegression
from sklearn.model_selection import train_test_split

X_tr, X_te, y_tr, y_te = train_test_split(
    X, y_true, test_size=0.2, random_state=42)

clf = LogisticRegression().fit(X_tr, y_tr)

print(f'Supervised accuracy: {clf.score(X_te, y_te):.2%}') # Uses labels
```

Supervised accuracy: 100.00%

```
# ■ Unsupervised approach: NO labels used at all ■
from sklearn.cluster import KMeans

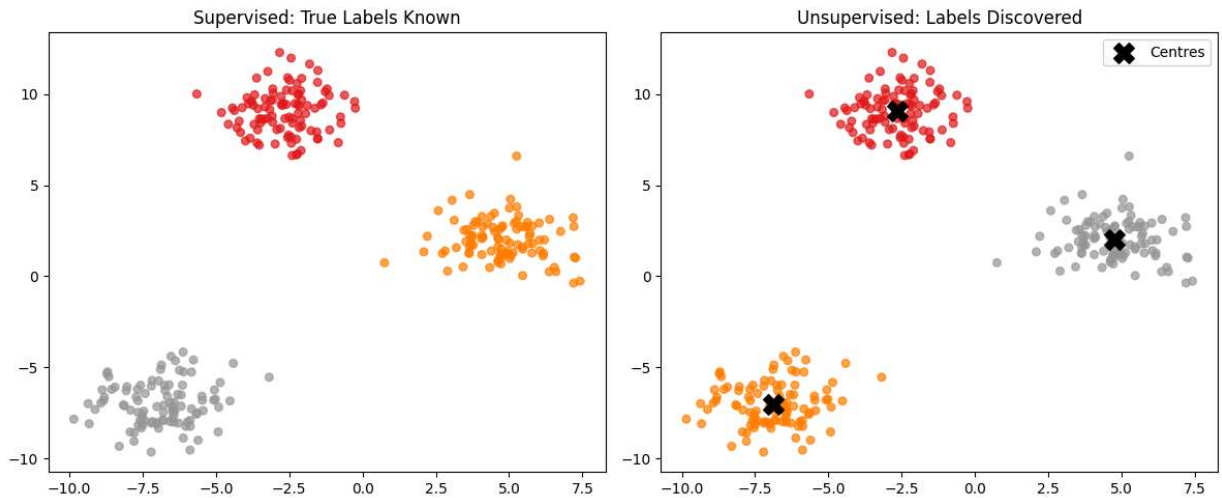
kmeans = KMeans(n_clusters=3, random_state=42, n_init=10)

cluster_labels = kmeans.fit_predict(X) # No y passed – only X!

print('Cluster assignments (first 10):', cluster_labels[:10])
# [0, 2, 1, 0, 2, ...] – but 0,1,2 are just cluster IDs, not class names
```

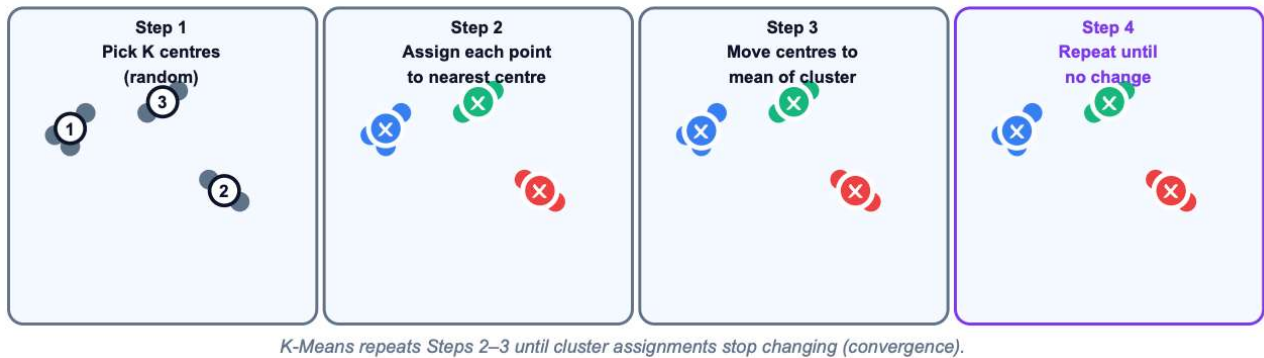
Cluster assignments (first 10): [1 1 0 2 1 2 0 2 0 0]

```
# ■■ Plot both side by side ■■
fig, (ax1, ax2) = plt.subplots(1, 2, figsize=(12, 5))
ax1.scatter(X[:,0], X[:,1], c=y_true, cmap='Set1', s=30, alpha=0.7)
ax1.set_title('Supervised: True Labels Known', fontsize=12)
ax2.scatter(X[:,0], X[:,1], c=cluster_labels, cmap='Set1', s=30, alpha=0.7)
ax2.scatter(*kmeans.cluster_centers_.T, c='black', s=200, marker='X', label='Centres')
ax2.set_title('Unsupervised: Labels Discovered', fontsize=12)
ax2.legend(); plt.tight_layout(); plt.show()
# The colours may differ but the shapes should match!
```



K-Means Clustering Algorithm

K-Means is the most popular clustering algorithm. Think of it as organising a messy drawer: you pick K spots to put dividers, assign every item to its nearest divider, then slide each divider to the middle of its group. Repeat until nothing moves.



The 4-Step Algorithm

K-Means is an unsupervised learning algorithm that groups similar data points into K clusters.

The algorithm repeatedly performs Assign → Update → Repeat until the clusters stop changing.

Step	What Happens	Simp
1. Initialize	Randomly place K cluster centers (centroids) in the data space	Pick K starting points randomly. Th
2. Assign	Each data point goes to its nearest centroid	Every point joins the cluster whose
3. Update	Move each centroid to the mean of all points assigned to that cluster	Recalculate the center of each clu
4. Repeat	Repeat Assign → Update until clusters stop changing	Continue adjusting clusters until th

MATHEMATICAL VIEW

Step	Formula
Initialize	Centroids = randomly choose K points from the dataset
Assign	Each point is assigned to the nearest centroid
Update	New centroid = average (mean) of all points in that cluster
Convergence	Stop when cluster assignments no longer change

K-MEANS LIMITATIONS

Limitation	Explanation
1. Need to Choose K in Advance	K-Means requires you to specify the number of clusters (K) before training. The al
2. Sensitive to Initial Centroids	Since centroids are chosen randomly at the beginning, different starting points car
3. Assumes Spherical Clusters	K-Means works best when clusters are roughly circular (2D) or spherical (multi-dir

K-Means Step by Step with Animation-Style Output

Build K-Means from scratch first, then use sklearn's version:

```
# ■ K-Means: Manual + sklearn ■
import numpy as np, matplotlib.pyplot as plt
from sklearn.datasets import make_blobs
from sklearn.cluster import KMeans
from sklearn.preprocessing import StandardScaler
```

```
# ■ Part 1: Manual K-Means ■
def manual_kmeans(X, k=3, max_iter=10, seed=42):
    rng = np.random.default_rng(seed)
    # Step 1: random initial centres
    centres = X[rng.choice(len(X), k, replace=False)]
    print(centres)
    for iteration in range(max_iter):
    # Step 2: Assign each point to nearest centre
        dists = np.linalg.norm(X[:, np.newaxis] - centres, axis=2) # (n, k)
        labels = np.argmin(dists, axis=1) # (n,)
    # Step 3: Move centres to mean of their cluster
        new_centres = np.array([X[labels == k_].mean(axis=0) for k_ in range(k)])
        print(f'Iter {iteration+1}: centre moved by {np.linalg.norm(new_centres-centres, axis=2)}')
        if np.allclose(centres, new_centres): print(' → Converged!'); break
        centres = new_centres
    return labels, centres
X, _ = make_blobs(n_samples=200, centers=3, random_state=42)
labels, centres = manual_kmeans(X, k=3)
```

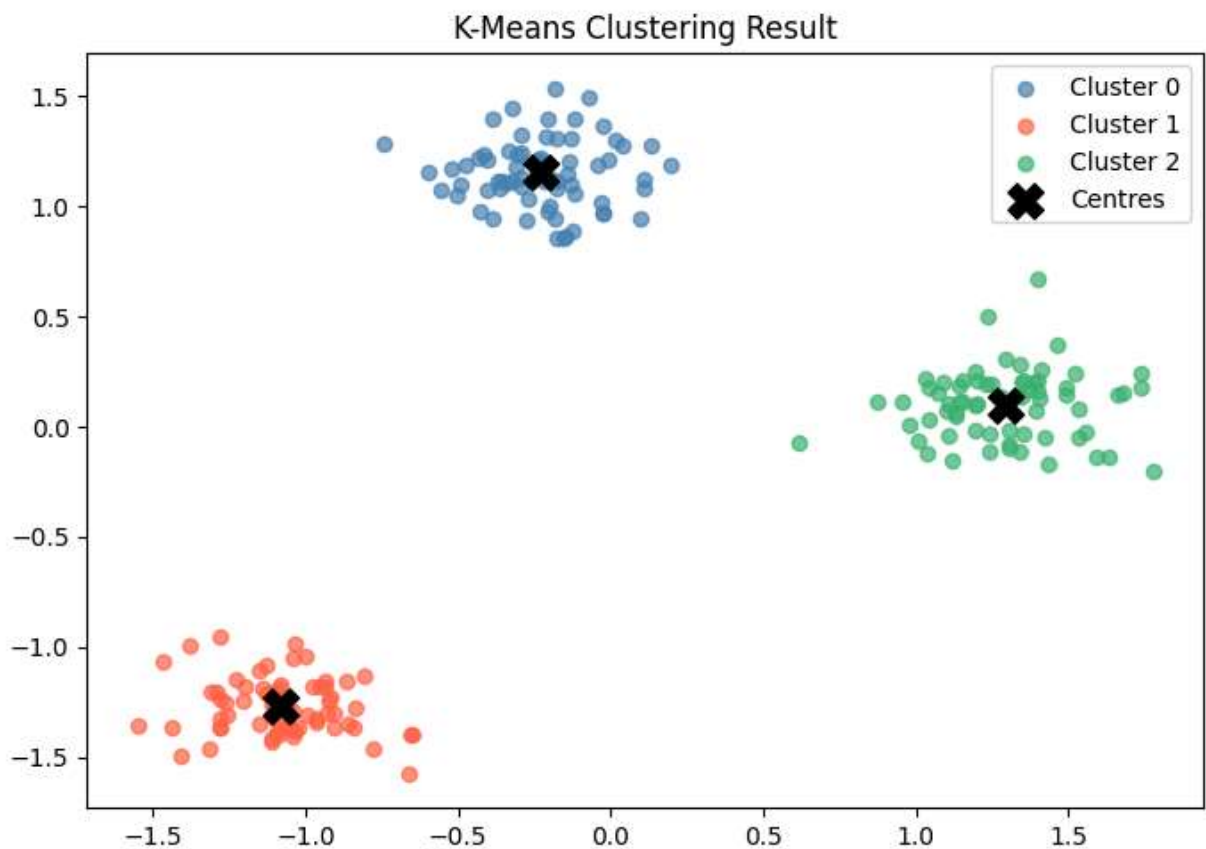
```
[[ 5.5938806  2.62456094]
 [-7.70184759 -6.63642238]
 [ 3.86705362  1.73635108]]
Iter 1: centre moved by 6.2785
Iter 2: centre moved by 3.6678
Iter 3: centre moved by 0.0000
 → Converged!
```

```
# ■ Part 2: sklearn KMeans (the production version) ■
scaler = StandardScaler() # Standardise BEFORE clustering
X_scaled = scaler.fit_transform(X) # K-Means uses distances → scale matters!
km = KMeans(
    n_clusters=3, # K – number of clusters
    init='k-means++', # Smart initialisation (better than random)
    n_init=10, # Run 10 times with different starts, keep best
    max_iter=300, # Max iterations per run
    random_state=42
)
km.fit(X_scaled)
print(f'Inertia : {km.inertia_:.2f}') # Total within-cluster variance
print(f'Iterations : {km.n_iter_}') # How many loops it took
print(f'Cluster sizes: {np.bincount(km.labels_)}') # e.g. [67 66 67]
```

```
Inertia : 12.18
Iterations : 2
```

Cluster sizes: [67 66 67]

```
# Visualise the result
plt.figure(figsize=(7,5))
colours = ['steelblue','tomato','mediumseagreen']
for k in range(3):
    mask = km.labels_ == k
    plt.scatter(X_scaled[mask,0], X_scaled[mask,1],
                c=colours[k], s=35, alpha=0.7, label=f'Cluster {k}')
plt.scatter(km.cluster_centers_[0], km.cluster_centers_[1],
            c='black', s=200, marker='X', zorder=5, label='Centres')
plt.title('K-Means Clustering Result'); plt.legend(); plt.tight_layout(); plt.s
```



Elbow Method — Finding the Right K

The Elbow Method helps determine the optimal number of clusters (K) by finding the point where increasing K no longer significantly reduces WCSS (Within in cluster sum of squares).

How do we know the right value of K?

The Elbow Method runs K-Means multiple times using different values of K:

$K = 1, 2, 3, 4, 5, \dots$

For each value of K, we calculate Inertia (WCSS).

What is Inertia?

Inertia measures:

"How close the data points are to their cluster center."

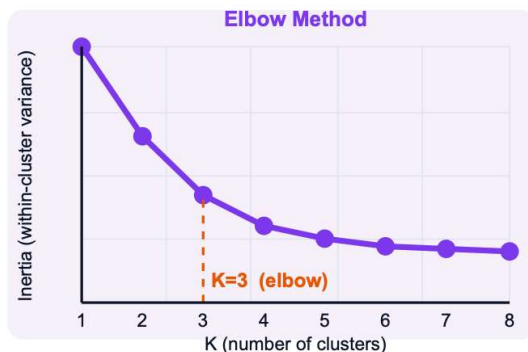
Lower Inertia → Better clustering Higher Inertia → Poor clustering

WCSS (Within-Cluster Sum of Squares)

WCSS is one of the most important concepts in K-Means Clustering.

How close the data points are to the center (centroid) of their cluster.

It calculates the total squared distance between each data point and its cluster centroid.



Inertia = sum of squared distances from each point to its cluster centre.

- K=1: one big cluster — very high inertia (messy)
- K=2: halved — big improvement
- K=3: another big jump — clear elbow ← **pick this**
- K=4, 5, 6... : tiny improvement — not worth it

The elbow is where adding one more cluster gives diminishing returns.

Elbow Method + Silhouette Score (Second Opinion)

Use both methods together — the elbow gives a range, silhouette score confirms the winner:

```
# ■ Elbow Method + Silhouette Score ■
from sklearn.cluster import KMeans
from sklearn.metrics import silhouette_score
from sklearn.preprocessing import StandardScaler
from sklearn.datasets import make_blobs
import matplotlib.pyplot as plt, numpy as np

#used to create artifical data
X,_= make_blobs(
```

```

n_samples=400,
centers=4,
cluster_std=1.5,
random_state=42)

#transforming large data to [-1,1]
X = StandardScaler().fit_transform(X)

inertias, silhouettes, k_range = [], [], range(2, 11)
for k in k_range:
    km = KMeans(n_clusters=k, n_init=10, random_state=42)
    print(k)
    km.fit(X)
    inertias.append(km.inertia_)
    sil = silhouette_score(X, km.labels_) # -1 to +1, higher = better
silhouettes.append(sil)
print(f'K={k:2d} Inertia={km.inertia_:8.2f} Silhouette={sil:.4f}')

```

```

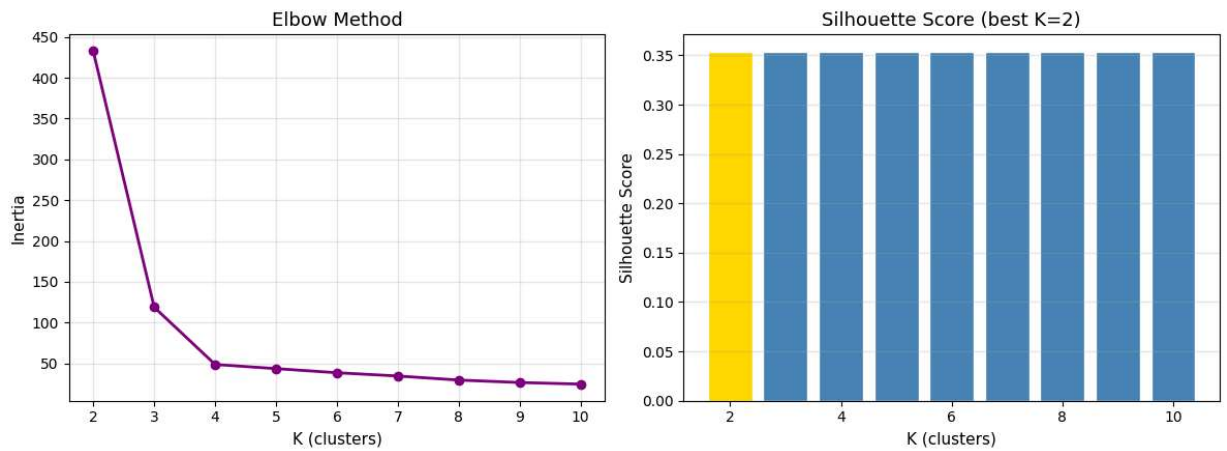
2
3
4
5
6
7
8
9
10
K=10 Inertia=    24.51 Silhouette=0.3534

```

```

# ■■■ Plot both ■■■
fig, (ax1, ax2) = plt.subplots(1, 2, figsize=(12, 4.5))
# Elbow plot
ax1.plot(k_range, inertias, 'o-', color='purple', lw=2)
ax1.set_xlabel('K (clusters)', fontsize=11); ax1.set_ylabel('Inertia', fontsize=11)
ax1.set_title('Elbow Method', fontsize=13); ax1.grid(alpha=0.3)
# Silhouette plot
best_k = k_range[silhouettes.index(max(silhouettes))]
bars = ax2.bar(k_range, silhouettes, color=['gold' if k==best_k else 'steelblue'
for k in k_range], edgecolor='white')
ax2.set_xlabel('K (clusters)', fontsize=11); ax2.set_ylabel('Silhouette Score',
    fontsize=11); ax2.set_title(f'Silhouette Score (best K={best_k})',
    fontsize=13); ax2.grid(alpha=0.3)
plt.tight_layout(); plt.show()
print(f'\nElbow suggests: look around K=3-5')
print(f'Silhouette says: best K = {best_k} (score={max(silhouettes):.4f})')

```

Elbow suggests: look around K=3-5
Silhouette says: best K = 2 (score=0.3534)

Visualising Clusters

Raw numbers tell you cluster assignments. Visualisation tells you what those clusters mean. Colour-coded scatter plots and cluster profile charts are the most powerful tools for understanding what each group represents

After applying K-Means Clustering, the algorithm gives us cluster labels such as:

Customer 1 → Cluster 0

Customer 2 → Cluster 1

Customer 3 → Cluster 2

These numbers tell us which cluster a data point belongs to, but they do not explain what the cluster actually represents.

This is where visualisation becomes important.

Raw Numbers vs Visualisation

Raw Numbers	Visualisation
Shows cluster IDs	Shows cluster meaning
Difficult to interpret	Easy to understand
Tells "which cluster"	Tells "why they belong together"
Useful for computation	Useful for business understanding

Cluster Visualisation Toolkit

Four ways to visualise and understand what your clusters mean:

```
# ■ Cluster Visualisation Toolkit ■
import pandas as pd, numpy as np, matplotlib.pyplot as plt
from sklearn.cluster import KMeans
from sklearn.preprocessing import StandardScaler
```

```
# Use mall customer dataset (from Lab 7) or simulate here
np.random.seed(42)
n=200

age = np.concatenate(
    [
        np.random.normal(25,5,60),
        np.random.normal(45,8,80),
        np.random.normal(60,6,60)]
)

income = np.concatenate(
    [np.random.normal(30,8,60),
     np.random.normal(60,12,80),
     np.random.normal(50,10,60)])

spending= np.concatenate(
    [np.random.normal(70,10,60),
     np.random.normal(50,15,80),
     np.random.normal(30,8,60)])

df = pd.DataFrame(
    {'Age':age,
     'Annual_Income':income,
     'Spending_Score':spending
    })

X = StandardScaler().fit_transform(df)

km = KMeans(n_clusters=3, n_init=10, random_state=42)
df['Cluster'] = km.fit_predict(X)
```

```
colors = ['#3B82F6', '#EF4444', '#10B981']
cluster_names = {0: 'Young Spenders', 1: 'Mid-Income Balanced', 2: 'Older Conservative'}
```

```
# ■ Plot 1: Scatter – Age vs Spending ■
fig, axes = plt.subplots(1, 3, figsize=(15, 4.5))
for c in range(3):
    mask = df['Cluster'] == c
    axes[0].scatter(df[mask]['Age'], df[mask]['Spending_Score'],
                    c=colors[c], s=40, alpha=0.7, label=cluster_names[c])
axes[0].set_xlabel('Age'); axes[0].set_ylabel('Spending Score')
axes[0].set_title('Age vs Spending'); axes[0].legend(fontsize=7)
# ■ Plot 2: Scatter – Income vs Spending ■
for c in range(3):
    mask = df['Cluster'] == c
    axes[1].scatter(df[mask]['Annual_Income'], df[mask]['Spending_Score'],
                    c=colors[c], s=40, alpha=0.7, label=cluster_names[c])
axes[1].set_xlabel('Annual Income'); axes[1].set_ylabel('Spending Score')
axes[1].set_title('Income vs Spending'); axes[1].legend(fontsize=7)
# ■ Plot 3: Cluster profiles (mean values per cluster) ■
profiles = df.groupby('Cluster')[['Age', 'Annual_Income', 'Spending_Score']].mean()
profiles.index = [cluster_names[i] for i in profiles.index]
profiles.T.plot(kind='bar', ax=axes[2], color=colors, edgecolor='white')
axes[2].set_title('Cluster Profiles (Mean Values)')
axes[2].set_xlabel('Feature'); axes[2].legend(fontsize=7)
axes[2].tick_params(axis='x', rotation=20)
plt.suptitle('Mall Customer Segments', fontsize=14, y=1.01)
plt.tight_layout(); plt.show()
```



Correlation Matrix & Heatmap

Correlation Matrix — Understanding Relationships Between Features

Before applying Clustering, Regression, or any Machine Learning model, it is important to understand how the features are related to each other.

A Correlation Matrix helps us measure the relationship between every pair of columns in a dataset.

What is Correlation?

Correlation tells us:

"When one feature changes, how does another feature change?"

For example:

As income increases, spending may increase.

As stress increases, sleep may decrease.

Correlation helps identify these relationships.

Pearson Correlation Coefficient

The most commonly used correlation measure is the Pearson Correlation Coefficient.

Its value ranges from:

-1 to +1

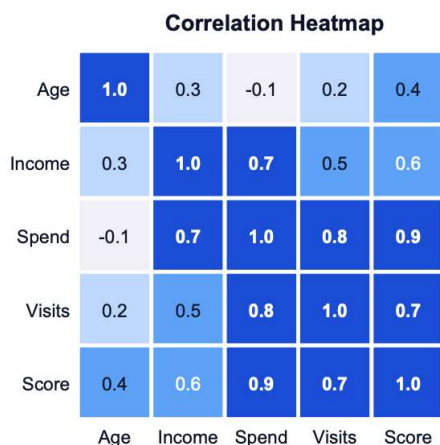
Where:

+1 → Perfect positive relationship

0 → No relationship

-1 → Perfect negative relationship

Correlation Value	Relationship	Meaning	Example
+0.7 to +1.0	Strong Positive	Both variables increase together	Income ↑ and Spending ↑
+0.3 to +0.7	Moderate Positive	One generally increases as the other increases	Age ↑ and Experience ↑
-0.1 to +0.1	No Correlation	No meaningful relationship	Shoe Size and IQ
-0.3 to -0.7	Moderate Negative	One increases while the other decreases	Stress ↑ and Sleep ↓
-0.7 to -1.0	Strong Negative	Strong opposite movement	Temperature ↑ and Heating



Reading the Heatmap

Dark blue = strong positive correlation (≥ 0.7)

Light blue = moderate positive

Grey/white = no correlation

Red = negative correlation

Diagonal is always 1.0 — a feature is perfectly correlated with itself.

High correlation between two features means they carry **redundant information** — you may only need one of them.

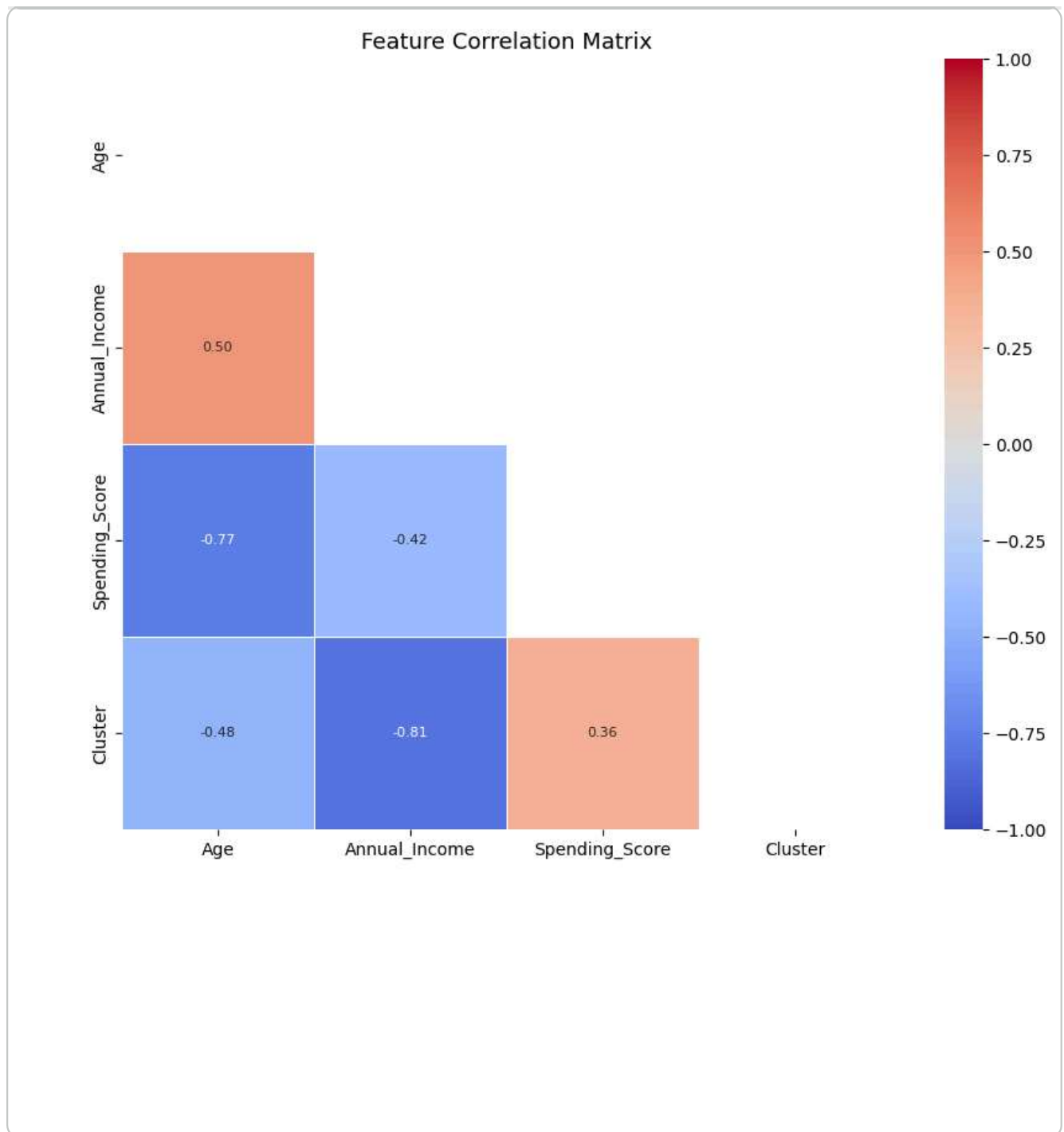
Correlation Matrix + Annotated Heatmap

```
# ■■■ Correlation Matrix + Heatmap ■■■
import pandas as pd, numpy as np, seaborn as sns, matplotlib.pyplot as plt
# Load or create a dataset
# df = pd.read_csv('diabetes.csv') # Using the 'df' created in the previous cel
```

```
# ■■■ Correlation matrix (numbers) ■■■
corr_matrix = df.corr(numeric_only=True) # Pearson by default
print('Correlation with Spending_Score:')
print(corr_matrix['Spending_Score'].sort_values(ascending=False).round(3))
```

```
Correlation with Spending_Score:
Spending_Score    1.000
Cluster           0.364
Annual_Income     -0.417
Age              -0.766
Name: Spending_Score, dtype: float64
```

```
# ■ Seaborn heatmap ■
plt.figure(figsize=(9, 7))
mask = np.triu(np.ones_like(corr_matrix, dtype=bool)) # hide upper triangle
sns.heatmap(
    corr_matrix,
    annot=True, # show values in each cell
    fmt='.2f', # 2 decimal places
    cmap='coolwarm', # blue=positive, red=negative
    mask=mask, # only show lower triangle
    vmin=-1, vmax=1, # fix colour scale
    linewidths=0.5,
    square=True,
    annot_kws={'size': 8}
)
plt.title('Feature Correlation Matrix', fontsize=13)
plt.tight_layout(); plt.show()
```



Outlier Detection with Boxplots & IQR

An outlier is a data point that is unusually far from the rest. Outliers can be errors (data entry mistake), or genuine extreme values (a millionaire in a survey). Either way, you need to detect and decide what to do with them before modelling

The IQR Method

The Interquartile Range (IQR) is the range of the middle 50% of data. Anything beyond

$$\text{IQR} = Q3 - Q1$$

$$\text{Lower fence} = Q1 - 1.5 \times \text{IQR}$$

$$\text{Upper fence} = Q3 + 1.5 \times \text{IQR}$$

Options for Handling Outliers

Option	When to Use	
Remove	When the value is clearly incorrect or impossible	Age = 300, S
Cap / Clip (Winsorizing)	When you want to reduce the effect of extreme values without deleting data	Replace Sala
Keep	When the outlier is a real and important observation	Fraud transa
Investigate	When the reason for the outlier is unknown	Unusually hig

Boxplots, IQR Detection, and Outlier Handling

```
# ■ Outlier Detection: Boxplots + IQR + Z-Score ■
import pandas as pd, numpy as np, matplotlib.pyplot as plt, seaborn as sns
#df = pd.read_csv('diabetes.csv') # Using the 'df' created in the previous cell
```

```
# ■ Part 1: Boxplots for all features ■
plt.figure(figsize=(14, 5))
df.drop('Cluster', axis=1).boxplot()
plt.title('Boxplots - Spot Outliers Visually', fontsize=13)
plt.xticks(rotation=30); plt.tight_layout(); plt.show()
# The dots beyond the whiskers = outliers
```

